

PREDICTING OIL RESERVOIR PERFORMANCE

Most reservoir engineering calculations involve the use of the material balance equation. Some of the most useful applications of the MBE require the concurrent use of fluid flow equations, e.g., Darcy's equation. Combining the two concepts would enable the engineer to predict the reservoir future production performance as a function of time. Without the fluid flow concepts, the MBE simply provides performance as a function of the average reservoir pressure. Prediction of the reservoir future performance is ordinarily performed in the following two phases:

Phase 1. Predicting cumulative hydrocarbon production as a function of declining reservoir pressure. This stage is accomplished without regard to:

- Actual number of wells
- Location of wells
- Production rate of individual wells
- Time required to deplete the reservoir

Phase 2. The second stage of prediction is the time-production phase. In these calculations, the reservoir performance data, as calculated from Phase 1, are correlated with time. It is necessary in this phase to account for the number of wells and the productivity of each well.

PHASE 1. RESERVOIR PERFORMANCE PREDICTION METHODS

The material balance equation in its various mathematical forms as presented in Chapter 11 is designed to provide estimates of the initial oil-in-place N , size of the gas cap m , and water influx W_e . To use the MBE to predict the reservoir future performance, it requires two additional relations:

- Equation of producing (instantaneous) gas-oil ratio
- Equation for relating saturations to cumulative oil production

These auxiliary mathematical expressions are presented as follows.

Instantaneous Gas-Oil Ratio

The produced gas-oil ratio (GOR) at any particular time is the ratio of the standard cubic feet of *total* gas being produced at any time to the stock-tank barrels of oil being produced at that same instant. Hence, the name *instantaneous gas-oil ratio*. Equation 6-54 in Chapter 6 describes the GOR mathematically by the following expression:

$$\text{GOR} = R_s + \left(\frac{k_{rg}}{k_{ro}} \right) \left(\frac{\mu_o B_o}{\mu_g B_g} \right) \quad (12-1)$$

where GOR = instantaneous gas-oil ratio, scf/STB

R_s = gas solubility, scf/STB

k_{rg} = relative permeability to gas

k_{ro} = relative permeability to oil

B_o = oil formation volume factor, bbl/STB

B_g = gas formation volume factor, bbl/scf

μ_o = oil viscosity, cp

μ_g = gas viscosity, cp

The instantaneous GOR equation is of fundamental importance in reservoir analysis. The importance of Equation 12-1 can appropriately be discussed in conjunction with Figures 12-1 and 12-2.

These illustrations show the history of the gas-oil ratio of a hypothetical depletion-drive reservoir that is typically characterized by the following points:

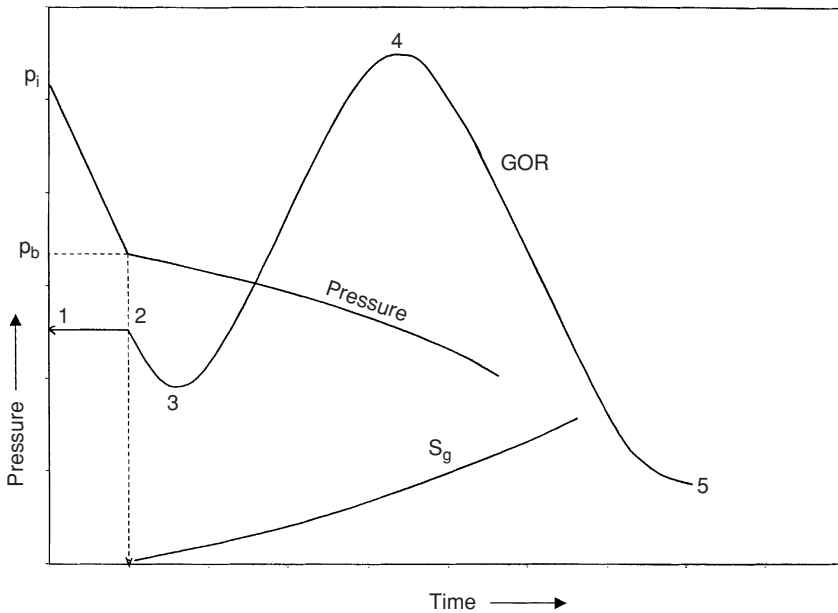


Figure 12-1. Characteristics of solution-gas-drive reservoirs.

Point 1. When the reservoir pressure p is above the bubble-point pressure p_b , there is no free gas in the formation, i.e., $k_{rg} = 0$, and therefore:

$$\text{GOR} = R_{si} = R_{sb} \quad (12-2)$$

The gas-oil ratio remains constant at R_{si} until the pressure reaches the bubble-point pressure at Point 2.

Point 2. As the reservoir pressure declines below p_b , the gas begins to evolve from solution and its saturation increases. This free gas, however, cannot flow until the gas saturation S_g reaches the critical gas saturation S_{gc} at Point 3. From Point 2 to Point 3, the instantaneous GOR is described by a decreasing gas solubility as:

$$\text{GOR} = R_s \quad (12-3)$$

Point 3. At Point 3, the free gas begins to flow with the oil and the values of GOR are progressively increasing with the declining reservoir pressure to Point 4. During this pressure decline period, the GOR is described by Equation 12-1, or:

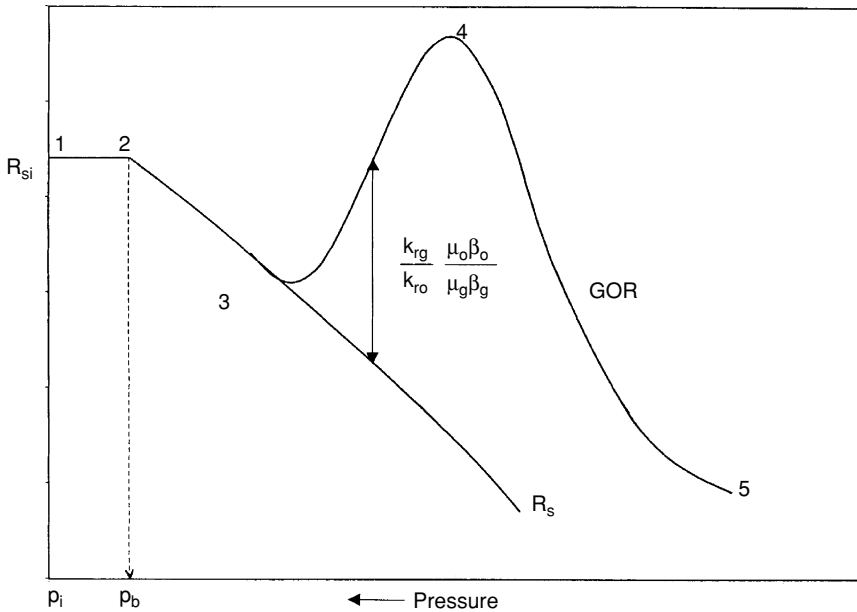


Figure 12-2. History of GOR and R_s for a solution-gas-drive reservoir.

$$\text{GOR} = R_s + \left(\frac{k_{rg}}{k_{ro}} \right) \left(\frac{\mu_o B_o}{\mu_g B_g} \right)$$

Point 4. At Point 4, the maximum GOR is reached due to the fact that the supply of gas has reached a maximum and marks the beginning of the *blow-down* period to Point 5.

Point 5. This point indicates that all the producible free gas has been produced and the GOR is essentially equal to the gas solubility and continues to Point 6.

There are three types of gas-oil ratios, all expressed in scf/STB, which must be clearly distinguished from each other. These are:

- Instantaneous GOR (defined by Equation 12-1)
- Solution GOR
- Cumulative GOR

The solution gas-oil ratio is a PVT property of the crude oil system. It is commonly referred to as *gas solubility* and denoted by R_s . It measures the tendency of the gas to dissolve in or evolve from the oil with changing pressures. It should be pointed out that as long as the evolved gas remains immobile, i.e., gas saturation S_g is less than the critical gas saturation, the instantaneous GOR is equal to the gas solubility, i.e.:

$$\text{GOR} = R_s$$

The cumulative gas-oil ratio R_p , as defined previously in the material balance equation, should be clearly distinguished from the producing (instantaneous) gas-oil ratio (GOR). The cumulative gas-oil ratio is defined as:

$$R_p = \frac{\text{cumulative (TOTAL) gas produced}}{\text{cumulative oil produced}}$$

or

$$R_p = \frac{G_p}{N_p} \quad (12-4)$$

where R_p = cumulative gas-oil ratio, scf/STB
 G_p = cumulative gas produced, scf
 N_p = cumulative oil produced, STB

The cumulative gas produced G_p is related to the instantaneous GOR and cumulative oil production by the expression:

$$G_p = \int_0^{N_p} (\text{GOR}) dN_p \quad (12-5)$$

Equation 12-5 simply indicates that the cumulative gas production at any time is essentially the area under the curve of the GOR versus N_p relationship, as shown in Figure 12-3.

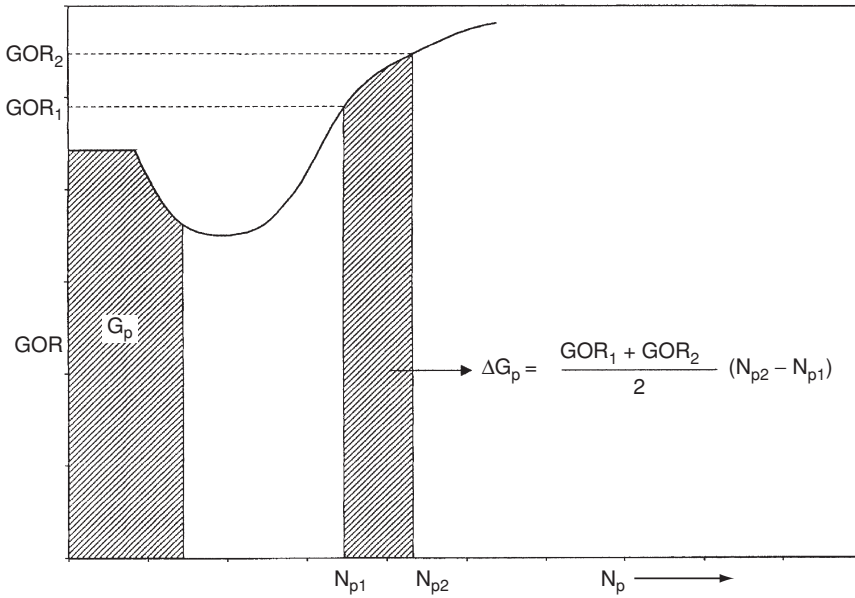


Figure 12-3. Relationship between GOR and G_p .

The incremental cumulative gas produced ΔG_p between N_{p1} , and N_{p2} is then given by:

$$\Delta G_p = \int_{N_{p1}}^{N_{p2}} (GOR) dN_p \quad (12-6)$$

The above integral can be approximated by using the trapezoidal rule, to give:

$$\Delta G_p = \left[\frac{(GOR)_1 + (GOR)_2}{2} \right] (N_{p2} - N_{p1})$$

or

$$\Delta G_p = (GOR)_{avg} \Delta N_p$$

Equation 12-5 can then be approximated as:

$$G_p = \sum_o (GOR)_{avg} \Delta N_p \quad (12-7)$$

Example 12-1

The following production data are available on a depletion-drive reservoir:

p psi	GOR scf/STB	N _p MMSTB
2925 (p _i)	1340	0
2600	1340	1.380
2400	1340	2.260
2100 (p _i)	1340	3.445
1800	1936	7.240
1500	3584	12.029
1200	6230	15.321

Calculate cumulative gas produced G_p and cumulative gas-oil ratio at each pressure.

Solution

Step 1. Construct the following table:

p psi	GOR scf/STB	(GOR) _{avg} scf/STB	N _p MMSTB	ΔN _p MMSTB	ΔG _p MMscf	G _p MMscf	R _p scf/STB
2925	1340	1340	0	0	0	0	—
2600	1340	1340	1.380	1.380	1849	1849	1340
2400	1340	1340	2.260	0.880	1179	3028	1340
2100	1340	1340	3.445	1.185	1588	4616	1340
1800	1936	1638	7.240	3.795	6216	10,832	1496
1500	3584	2760	12.029	4.789	13,618	24,450	2033
1200	6230	4907	15.321	3.292	16,154	40,604	2650

The Reservoir Saturation Equations

The saturation of a fluid (gas, oil, or water) in the reservoir is defined as the volume of the fluid divided by the pore volume, or:

$$S_o = \frac{\text{oil volume}}{\text{pore volume}} \quad (12-8)$$

$$S_w = \frac{\text{water volume}}{\text{pore volume}} \quad (12-9)$$

$$S_g = \frac{\text{gas volume}}{\text{pore volume}} \quad (12-10)$$

$$S_o + S_w + S_g = 1.0 \quad (12-11)$$

Consider a volumetric oil reservoir with no gas cap that contains N stock-tank barrels of oil at the initial reservoir pressure p_i . Assuming no water influx gives:

$$S_{oi} = 1 - S_{wi}$$

where the subscript i indicates initial reservoir condition. From the definition of oil saturation:

$$1 - S_{wi} = \frac{N B_{oi}}{\text{pore volume}}$$

or

$$\text{pore volume} = \frac{N B_{oi}}{1 - S_{wi}} \quad (12-12)$$

If the reservoir has produced N_p stock-tank barrels of oil, the remaining oil volume is given by:

$$\text{remaining oil volume} = (N - N_p) B_o \quad (12-13)$$

Substituting Equations 12-13 and 12-12 into Equation 12-8 gives:

$$S_o = \frac{(N - N_p) B_o}{\left(\frac{N B_{oi}}{1 - S_{wi}} \right)} \quad (12-14)$$

or

$$S_o = (1 - S_{wi}) \left(1 - \frac{N_p}{N} \right) \frac{B_o}{B_{oi}} \quad (12-15)$$

$$S_g = 1 - S_o - S_{wi} \quad (12-16)$$

Example 12-2

A volumetric solution-gas-drive reservoir has an initial water saturation of 20%. The initial oil formation volume factor is reported at 1.5 bbl/STB. When 10% of the initial oil was produced, the value of B_o decreased to 1.38. Calculate the oil saturation and gas saturation.

Solution

From Equation 12-5

$$S_o = (1 - 0.2)(1 - 0.1) \left(\frac{1.38}{1.50} \right) = 0.662$$

$$S_g = 1 - 0.662 - 0.20 = 0.138$$

It should be pointed out that the values of the relative permeability ratio k_{rg}/k_{ro} as a function of oil saturation can be generated by using the actual field production as expressed in terms of N_p , GOR, and PVT data. The proposed methodology involves the following steps:

Step 1. Given the actual field cumulative oil production N_p and the PVT data as a function of pressure, calculate the oil and gas saturations from Equations 12-15 and 12-16, i.e.:

$$S_o = (1 - S_{wi}) \left(1 - \frac{N_p}{N} \right) \frac{B_o}{B_{oi}}$$

$$S_g = 1 - S_o - S_{wi}$$

Step 2. Using the actual field instantaneous GORs, solve Equation 12-1 for the relative permeability ratio as:

$$\frac{k_{rg}}{k_{ro}} = (GOR - R_s) \left(\frac{\mu_g B_g}{\mu_o B_o} \right) \quad (12-17)$$

Step 3. Plot (k_{rg}/k_{ro}) versus S_o on a semilog paper.

Equation 12-15 suggests that all the remaining oil saturation be distributed uniformly throughout the reservoir. If water influx, gas-cap expansion, or gas-cap shrinking has occurred, the oil saturation equation, i.e., Equation 12-15, must be adjusted to account for oil trapped in the invaded regions.

Oil saturation adjustment for water influx

The proposed oil saturation adjustment methodology is illustrated in Figure 12-4 and described by the following steps:

Step 1. Calculate the pore volume in the water-invaded region, as:

$$W_e - W_p B_w = (P.V)_{water} (1 - S_{wi} - S_{orw})$$

Solving for the pore volume of water-invaded zone $(P.V)_{water}$ gives:

$$(P.V)_{water} = \frac{W_e - W_p B_w}{1 - S_{wi} - S_{orw}} \quad (12-18)$$

where $(P.V)_{water}$ = pore volume in water-invaded zone, bbl
 S_{orw} = residual oil saturated in the imbibition water-oil system

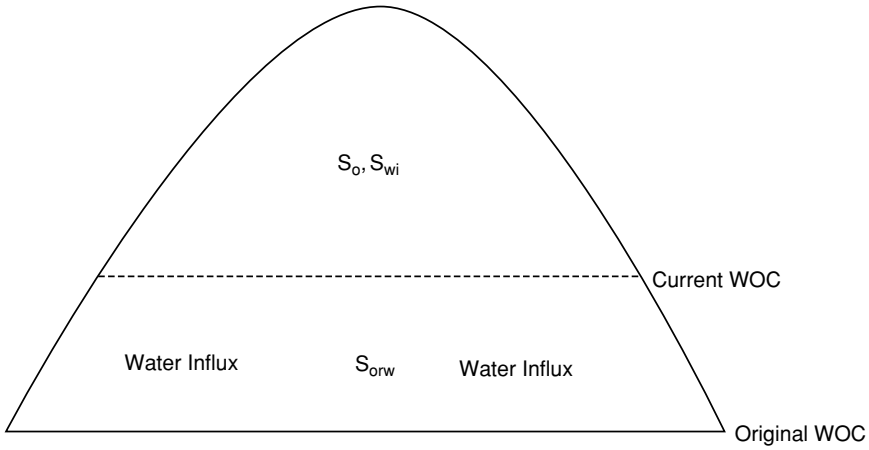


Figure 12-4. Oil saturation adjusted for water influx.

Step 2. Calculate oil volume in the water-invaded zone, or:

$$\text{volume of oil} = (\text{P.V})_{\text{water}} S_{\text{orw}} \quad (12-19)$$

Step 3. Adjust Equation 12-14 to account for the trapped oil by using Equations 12-18 and 12-19:

$$S_o = \frac{(N - N_p) B_o - \left[\frac{W_e - W_p B_w}{1 - S_{wi} - S_{orw}} \right] S_{orw}}{\left(\frac{N B_{oi}}{1 - S_{wi}} \right) - \left[\frac{W_e - W_p B_w}{1 - S_{wi} - S_{orw}} \right]} \quad (12-20)$$

Oil saturation adjustment for gas-cap expansion

The oil saturation adjustment procedure is illustrated in Figure 12-5 and summarized below:

Step 1. Assuming no gas is produced from the gas cap, calculate the net expansion of the gas cap, from:

$$\text{Expansion of the gas cap} = m N B_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right) \quad (12-21)$$

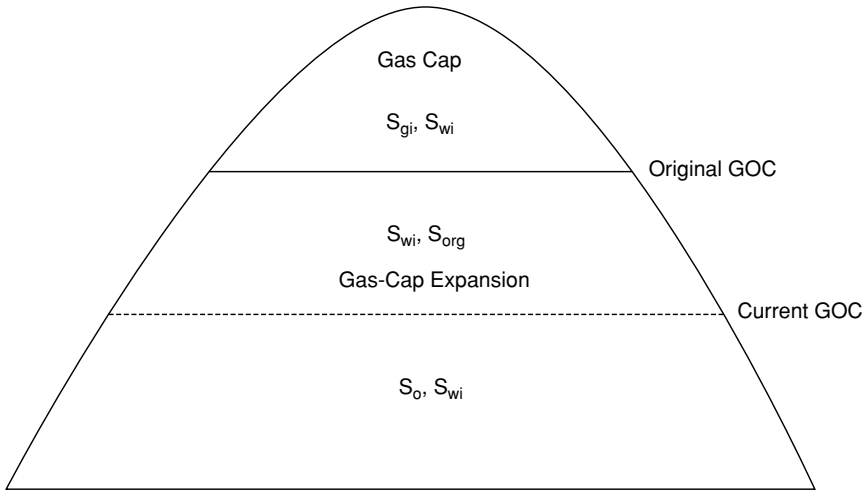


Figure 12-5. Oil saturation adjustment for gas-cap expansion.

Step 2. Calculate the pore volume of the gas-invaded zone, $(P.V)_{\text{gas}}$, by solving the following simple material balance:

$$mNB_{\text{oi}} \left(\frac{B_g}{B_{\text{gi}}} - 1 \right) = (P.V)_{\text{gas}} (1 - S_{\text{wi}} - S_{\text{org}})$$

or

$$(P.V)_{\text{gas}} = \frac{mNB_{\text{oi}} \left(\frac{B_g}{B_{\text{gi}}} - 1 \right)}{1 - S_{\text{wi}} - S_{\text{org}}} \quad (12-22)$$

where $(P.V)_{\text{gas}}$ = pore volume of the gas-invaded zone
 S_{org} = residual oil saturation in gas-oil system

Step 3. Calculate the volume of oil in the gas-invaded zone.

$$\text{oil volume} = (P.V)_{\text{gas}} S_{\text{org}} \quad (12-23)$$

Step 4. Adjust Equation 12-14 to account for the trapped oil in the gas expansion zone by using Equations 12-22 and 12-23, to give:

$$S_o = \frac{(N - N_p)B_o - \left[\frac{mNB_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right)}{1 - S_{wi} - S_{org}} \right] S_{org}}{\left(\frac{NB_{oi}}{1 - S_{wi}} \right) - \left[\frac{mNB_{oi}}{1 - S_{wi} - S_{org}} \right] \left(\frac{B_g}{B_{gi}} - 1 \right)} \quad (12-24)$$

Oil saturation adjustment for combination drive

For a combination-drive reservoir, i.e., water influx and gas cap, the oil saturation equation as given by Equation 12-14 can be adjusted to account for both driving mechanisms, as:

$$S_o = \frac{(N - N_p)B_o - \left[\frac{mNB_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right) S_{org}}{1 - S_{wi} - S_{org}} + \frac{(W_e - W_p)S_{orw}}{1 - S_{wi} - S_{orw}} \right]}{\frac{NB_{oi}}{1 - S_{wi}} - \left[\frac{mNB_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right)}{1 - S_{wi} - S_{org}} + \frac{W_e - W_p}{1 - S_{wi} - S_{orw}} \frac{B_w}{B_o} \right]} \quad (12-25)$$

Oil saturation adjustment for shrinking gas cap

Cole (1969) points out that the control of the gas-cap size is very often a reliable guide to the efficiency of reservoir operations. A shrinking gas cap will cause the loss of a substantial amount of oil, which might otherwise be recovered. Normally, there is little or no oil saturation in the gas cap, and if the oil migrates into the original gas zone, there will necessarily

be some residual oil saturation remaining in this portion of the gas cap at abandonment. The magnitude of this loss may be quite large, depending upon the:

- Area of the gas-oil contact
- Rate of gas-cap shrinkage
- Relative permeability characteristics
- Vertical permeability

A shrinking gas cap can be controlled by either shutting in wells that are producing large quantities of gas-cap gas or by returning some of the produced gas back to the gas cap portion of the reservoir. In many cases, the shrinkage cannot be completely eliminated by shutting in wells, as there is a practical limit to the number of wells that can be shut-in. The amount of oil lost by the shrinking gas cap can be very well the engineer's most important economic justification for the installation of gas return facilities.

The difference between the original volume of the gas cap and the volume occupied by the gas cap at any subsequent time is a measure of the volume of oil that has migrated into the gas cap. If the size of the original gas cap is $m N B_{oi}$, then the expansion of the original free gas resulting from reducing the pressure from p_i to p is:

$$\text{Expansion of the original gas cap} = m N B_{oi} [(B_g/B_{gi}) - 1]$$

where $m N B_{oi}$ = original gas-cap volume, bbl

B_g = gas FVF, bbl/scf

If the gas cap is shrinking, then the volume of the produced gas must be larger than the gas-cap expansion. All of the oil that moves into the gas cap will not be lost, as this oil will also be subject to the various driving mechanisms. Assuming no original oil saturation in the gas zone, the oil that will be lost is essentially the residual oil saturation remaining at abandonment. If the cumulative gas production from the gas cap is G_{pc} scf, the volume of the gas-cap shrinkage as expressed in barrels is equal to:

$$\text{Gas-cap shrinkage} = G_{pc} B_g - m N B_{oi} [(B_g/B_{gi}) - 1]$$

From the volumetric equation:

$$G_{pc} B_g - m N B_{oi} [(B_g/B_{gi}) - 1] = 7758 A h \phi (1 - S_{wi} - S_{gr})$$

where A = average cross-sectional area of the gas-oil contact, acres

h = average change in depth of the gas-oil contact, feet

S_{gr} = residual gas saturation in the shrinking zone

The volume of oil lost as a result of oil migration to the gas cap can also be calculated from the volumetric equation as follows:

$$\text{Oil lost} = 7758 A h \phi S_{org}/B_{oa}$$

where S_{org} = residual oil saturation in the gas-cap shrinking zone

B_{oa} = oil FVF at abandonment

Combining the above relationships and eliminating the term $7,758 A h \phi$, give the following expression for estimating the volume of oil in barrels lost in the gas cap:

$$\text{Oil lost} = \frac{\left[G_{pc} B_g - m N B_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right) \right] S_{org}}{(1 - S_{wi} - S_{gr}) B_{oa}}$$

where G_{pc} = cumulative gas production for the gas cap, scf

B_g = gas FVF, bbl/scf

All the methodologies that have been developed to predict the future reservoir performance are essentially based on employing and combining the above relationships that include the:

- MBE
- Saturation equations
- Instantaneous GOR
- Equation relating the cumulative gas-oil ratio to the instantaneous GOR

Using the above information, it is possible to predict the field primary recovery performance with declining reservoir pressure. There are three

methodologies that are widely used in the petroleum industry to perform a reservoir study. These are:

- Tracy's method
- Muskat's method
- Turner's method

These methods yield essentially the same results when small intervals of pressure or time are used. The methods can be used to predict the performance of a reservoir under any driving mechanism, including:

- Solution-gas drive
- Gas-cap drive
- Water drive
- Combination drive

The practical use of all the techniques is illustrated in predicting the primary recovery performance of a volumetric solution-gas-drive reservoir. Using the appropriate saturation equation, e.g., Equation 12-20 for a water-drive reservoir, any of the available reservoir prediction techniques could be applied to other reservoirs operating under different driving mechanisms.

The following two cases of the solution-gas-drive reservoir are considered:

- Undersaturated-oil reservoirs
- Saturated-oil reservoirs

Undersaturated-Oil Reservoirs

When the reservoir pressure is above the bubble-point pressure of the crude oil system, the reservoir is considered undersaturated. The general material balance is expressed in Chapter 11 by Equation 11-15.

$$N = \frac{N_p [B_o + (R_p - R_s) B_g] - (W_e - W_p B_w) - G_{inj} B_{ginj} - W_{inj} B_{wi}}{(B_o - B_{oi}) + (R_{si} - R_s) B_g + m B_{oi} \left[\frac{B_g}{B_{gi}} - 1 \right] + B_{oi} (1 + m) \left[\frac{S_{wi} c_w + c_f}{1 - S_{wi}} \right] \Delta p}$$

For a volumetric undersaturated reservoir with no fluid injection, the following conditions are observed:

$$\begin{aligned}
 m &= 0 \\
 W_e &= 0 \\
 R_s &= R_{si} = R_p
 \end{aligned}$$

Imposing the above conditions on the MBE reduces the equation to the following simplified form:

$$N = \frac{N_p B_o}{(B_o - B_{oi}) + B_{oi} \left[\frac{S_{wi} c_w + c_f}{1 - S_{wi}} \right] \Delta p} \quad (12-26)$$

with

$$\Delta p = p_i - p$$

where p_i = initial reservoir pressure
 p = current reservoir pressure

Hawkins (1955) introduced the oil compressibility c_o into the MBE to further simplify the equation. The oil compressed is defined in Chapter 2 by:

$$c_o = \frac{1}{B_{oi}} \frac{B_o - B_{oi}}{\Delta p}$$

Rearranging gives:

$$B_o - B_{oi} = c_o B_{oi} \Delta p$$

Combining the above expression with Equation 12-26 gives:

$$N = \frac{N_p B_o}{c_o B_{oi} \Delta p + B_{oi} \left[\frac{S_{wi} c_w + c_f}{1 - S_{wi}} \right] \Delta p} \quad (12-27)$$

The denominator of the above equation can be written as:

$$B_{oi} \left[c_o + \frac{S_{wi} c_w}{1 - S_{wi}} + \frac{c_f}{1 - S_{wi}} \right] \Delta p \quad (12-28)$$

Since there are only two fluids in the reservoir, i.e., oil and water, then:

$$S_{oi} + S_{wi} = 1$$

Equation 12-28 can then be expressed as:

$$B_{oi} \left[\frac{S_{oi} c_o + S_{wi} c_w + c_f}{1 - S_{wi}} \right] \Delta p$$

The term between the two brackets is called the effective compressibility and defined by Hawkins (1955) as:

$$c_e = \frac{S_{oi} c_o + S_{wi} c_w + c_f}{1 - S_{wi}} \quad (12-29)$$

Combining Equations 12-27, 12-28, and 12-29, the MBE above the bubble-point pressure becomes:

$$N = \frac{N_p B_o}{B_{oi} c_e \Delta p} = \frac{N_p B_o}{B_{oi} c_e (P_i - P)} \quad (12-30)$$

Equation 12-30 can be expressed as an equation of a straight line by:

$$P = P_i - \left[\frac{1}{N B_{oi} c_e} \right] N_p B_o \quad (12-31)$$

Figure 12-6 indicates that the reservoir pressure will decrease linearly with cumulative reservoir voidage $N_p B_o$.

Rearranging Equation 12-31 and solving for the cumulative oil production N_p gives:

$$N_p = N c_e \left(\frac{B_o}{B_{oi}} \right) \Delta p \quad (12-32)$$

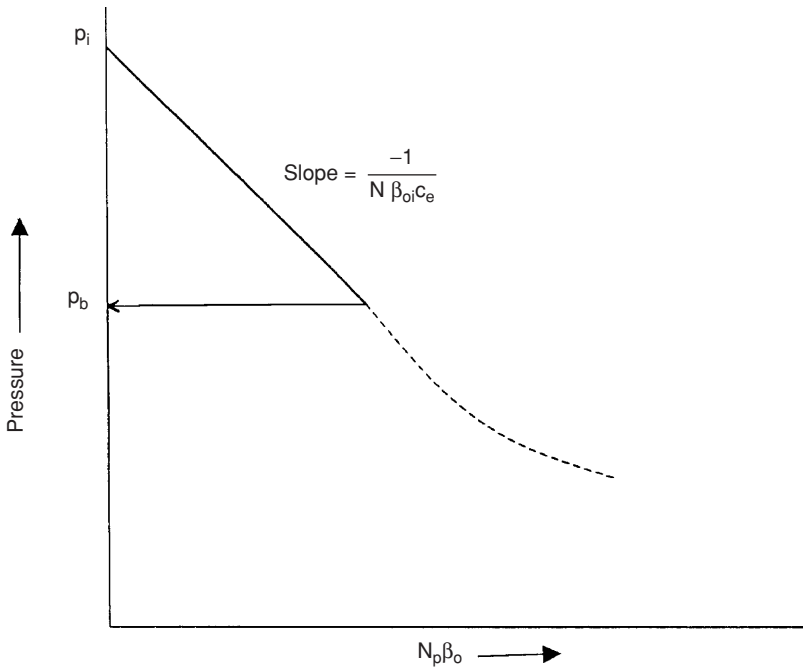


Figure 12-6. Pressure voidage relationship.

The calculation of future reservoir production, therefore, does not require a trial-and-error procedure, but can be obtained directly from the above expression.

Example 12-3

The following data are available on a volumetric undersaturated-oil reservoir:

$p_i = 4000 \text{ psi}$	$p_b = 3000 \text{ psi}$	$N = 85 \text{ MMSTB}$
$c_f = 5 \times 10^{-6} \text{ psi}^{-1}$	$c_o = 15 \times 10^{-6} \text{ psi}^{-1}$	$c_w = 3 \times 10^{-6} \text{ psi}^{-1}$
$S_{wi} = 30\%$	$B_{oi} = 1.40 \text{ bbl/STB}$	

Estimate cumulative oil production when the reservoir pressure drops to 3,500 psi. The oil formation volume factor at 3,500 psi is 1.414 bbl/STB.

Solution

Step 1. Determine the effective compressibility from Equation 12-29.

$$c_e = \frac{(0.7)(15 \times 10^{-6}) + (0.3)(3 \times 10^{-6}) + 5 \times 10^{-6}}{1 - 0.3}$$

$$= 23.43 \times 10^{-6} \text{ psi}^{-1}$$

Step 2. Estimate N_p from Equation 12-32.

$$N_p = (85 \times 10^{-6})(23.43 \times 10^{-6}) \left(\frac{1.411}{1.400} \right) (4000 - 3500)$$

$$= 985.18 \text{ MSTB}$$

Saturated-Oil Reservoirs

If the reservoir originally exists at its bubble-point pressure, the reservoir is referred to as a saturated-oil reservoir. This is considered as the second type of the solution-gas-drive reservoir. As the reservoir pressure declines below the bubble-point, the gas begins to evolve from solution. The general MBE may be simplified by assuming that the expansion of the gas is much greater than the expansion of rock and initial water and, therefore, can be neglected. For a volumetric and saturated-oil reservoir with no fluid injection, the MBE can be expressed by:

$$N = \frac{N_p B_o + (G_p - N_p R_s) B_g}{(B_o - B_{oi}) + (R_{si} - R_s) B_g} \quad (12-33)$$

The above material balance equation contains two unknowns, which are:

- Cumulative oil production N_p
- Cumulative gas production G_p

The following reservoir and PVT data must be available in order to predict the primary recovery performance of a depletion-drive reservoir in terms of N_p and G_p :

a. Initial oil-in-place N

Generally, the volumetric estimate of in-place oil is used in calculating the performance. Where there is sufficient solution-gas-drive history,

however, this estimate may be checked by calculating a material-balance estimate.

b. Hydrocarbon PVT data

Since differential gas liberation is assumed to best represent the conditions in the reservoir, differential laboratory PVT data should be used in reservoir material balance. The flash PVT data are then used to convert from reservoir conditions to stock-tank conditions.

If laboratory data are not available, reasonable estimates may sometimes be obtained from published correlations. If differential data are not available, the flash data may be used instead; however, this may result in large errors for high-solubility crude oils.

c. Initial fluid saturations

Initial fluid saturations obtained from a laboratory analysis of core data are preferred; however, if these are not available, estimates in some cases may be obtained from a well-log analysis or may be obtained from other reservoirs in the same or similar formations.

d. Relative permeability data

Generally, laboratory-determined k_g/k_o and k_{ro} data are averaged to obtain a single representative set for the reservoir. If laboratory data are not available, estimates in some cases may be obtained from other reservoirs in the same or similar formations.

Where there is sufficient solution-gas-drive history for the reservoir, calculate (k_{rg}/k_{ro}) values versus saturation from Equations 12-15 and 12-17, i.e.:

$$S_o = (1 - S_{wi}) (1 - N_p/N) (B_o/B_{oi})$$

$$k_{rg}/k_{ro} = (GOR - R_s) (\mu_g B_g/\mu_o B_o)$$

The above results should be compared with the averaged laboratory relative permeability data. This may indicate a needed adjustment in the early data and possibly an adjustment in the overall data.

All the techniques that are used to predict the future performance of a reservoir are based on combining the appropriate MBE with the instantaneous GOR using the proper saturation equation. The calculations are repeated at a series of assumed reservoir pressure drops. These calculations are usually based on one stock-tank barrel of oil-in-place at the

bubble-point pressure, i.e., $N = 1$. This avoids carrying large numbers in the calculation procedure and permits calculations to be made on the basis of the fractional recovery of initial oil-in-place.

There are several widely used techniques that were specifically developed to predict the performance of solution-gas-drive reservoirs, including:

- Tracy's method
- Muskat's method
- Tarner's method

These methodologies are presented below.

Tracy's Method

Tracy (1955) suggests that the general material balance equation can be rearranged and expressed in terms of three functions of PVT variables. Tracy's arrangement is given in Chapter 11 by Equation 11-53 and is repeated here for convenience:

$$N = N_p \Phi_o + G_p \Phi_g + (W_p B_w - W_e) \Phi_w \quad (12-34)$$

where Φ_o , Φ_g , and Φ_w are considered PVT-related properties that are functions of pressure and defined by:

$$\Phi_o = \frac{B_o - R_s B_g}{\text{Den}}$$

$$\Phi_g = \frac{B_g}{\text{Den}}$$

$$\Phi_w = \frac{1}{\text{Den}}$$

with

$$\text{Den} = (B_o - B_{oi}) + (R_{si} - R_s) B_g + m B_{oi} \left[\frac{B_g}{B_{gi}} - 1 \right] \quad (12-35)$$

For a solution-gas-drive reservoir, Equations 12-34 and 12-35 are reduced to the following expressions, respectively:

$$N = N_p \Phi_o + G_p \Phi_g \quad (12-36)$$

and

$$\text{Den} = (B_o - B_{oi}) + (R_{si} - R_s) B_g \quad (12-37)$$

Tracy's calculations are performed in series of pressure drops that proceed from a known reservoir condition at the previous reservoir pressure p^* to the new assumed lower pressure p . The calculated results at the new reservoir pressure become "known" at the next assumed lower pressure.

In progressing from the conditions at any pressure p^* to the lower reservoir pressure p , consider that the incremental oil and gas production are ΔN_p and ΔG_p , or:

$$N_p = N_p^* + \Delta N_p \quad (12-38)$$

$$G_p = G_p^* + \Delta G_p \quad (12-39)$$

where N_p^*, G_p^* = "known" cumulative oil and gas production at previous pressure level p^*

N_p, G_p = "unknown" cumulative oil and gas at new pressure level p

Replacing N_p and G_p in Equation 12-36 with those of Equations 12-38 and 12-39 gives:

$$N = (N_p^* + \Delta N_p) \Phi_o + (G_p^* + \Delta G_p) \Phi_g \quad (12-40)$$

Define the average instantaneous GOR between the two pressures p^* and p by:

$$(\text{GOR})_{\text{avg}} = \frac{\text{GOR}^* + \text{GOR}}{2} \quad (12-41)$$

The incremental cumulative gas production ΔG_p can be approximated by Equation 12-7 as:

$$\Delta G_p = (\text{GOR})_{\text{avg}} \Delta N_p \quad (12-42)$$

Replacing ΔG_p in Equation 12-40 with that of 12-41 gives:

$$N = [N_p^* + \Delta N_p] \Phi_o + [G_p^* + \Delta N_p (GOR)_{avg}] \Phi_g \quad (12-43)$$

If Equation 12-43 is expressed for $N = 1$, the cumulative oil production N_p and cumulative gas production G_p become fractions of initial oil-in-place. Rearranging Equation 12-43 gives:

$$\Delta N_p = \frac{1 - (N_p^* \Phi_o + G_p^* \Phi_g)}{\Phi_o + (GOR)_{avg} \Phi_g} \quad (12-44)$$

Equation 12-44 shows that there are essentially two unknowns, the incremental cumulative oil production ΔN_p and the average gas-oil ratio $(GOR)_{avg}$.

Tracy suggested the following alternative technique for solving Equation 12-44.

Step 1. Select an average reservoir pressure p .

Step 2. Calculate the values of the PVT functions Φ_o and Φ_g .

Step 3. Estimate the GOR at p .

Step 4. Calculate the average instantaneous GOR $(GOR)_{avg} = (GOR^* + GOR)/2$.

Step 5. Calculate the incremental cumulative oil production ΔN_p from Equation 12-44 as:

$$\Delta N_p = \frac{1 - (N_p^* \Phi_o + G_p^* \Phi_g)}{\Phi_o + (GOR)_{avg} \Phi_g}$$

Step 6. Calculate cumulative oil production N_p :

$$N_p = N_p^* + \Delta N_p$$

Step 7. Calculate the oil and gas saturations at selected average reservoir pressure by using Equations 12-15 and 12-16, as:

$$S_o = (1 - S_{wi}) (1 - N_p) (B_o/B_{oi})$$

$$S_g = 1 - S_o - S_{wi}$$

Step 8. Obtain the relative permeability ratio k_{rg}/k_{ro} at S_g .

Step 9. Calculate the instantaneous GOR from Equation 12-1.

$$GOR = R_s + (k_{rg}/k_{ro}) (\mu_o B_o/\mu_g B_g)$$

Step 10. Compare the estimated GOR in Step 3 with the calculated GOR in Step 9. If the values are within the acceptable tolerance, proceed to the next step. If not within the tolerance, set the estimated GOR equal to the calculated GOR and repeat the calculations from Step 3.

Step 11. Calculate the cumulative gas production.

$$G_p = G_p^* + \Delta N_p (GOR)_{avg}$$

Step 12. Since results of the calculations are based on 1 STB of oil initially in place, a final check on the accuracy of the prediction should be made on the MBE, or:

$$N_p \Phi_o + G_p \Phi_g = 1 \pm \text{tolerance}$$

Step 13. Repeat from Step 1.

As the calculation progresses, a plot of GOR versus pressure can be maintained and extrapolated as an aid in estimating GOR at each new pressure.

Example 12-4¹

The following PVT data characterize a solution-gas-drive reservoir.

¹The example data and solution are given by Economides, M., Hill, A., and Economides, C., *Petroleum Production System*, Prentice Hall Petroleum Engineering series, 1994.

The relative permeability data are shown in Figure 12-7.

p psi	B _o bbl/STB	B _g bbl/scf	R _s scf/STB
4350	1.43	6.9×10^{-4}	840
4150	1.420	7.1×10^{-4}	820
3950	1.395	7.4×10^{-4}	770
3750	1.380	7.8×10^{-4}	730
3550	1.360	8.1×10^{-4}	680
3350	1.345	8.5×10^{-4}	640

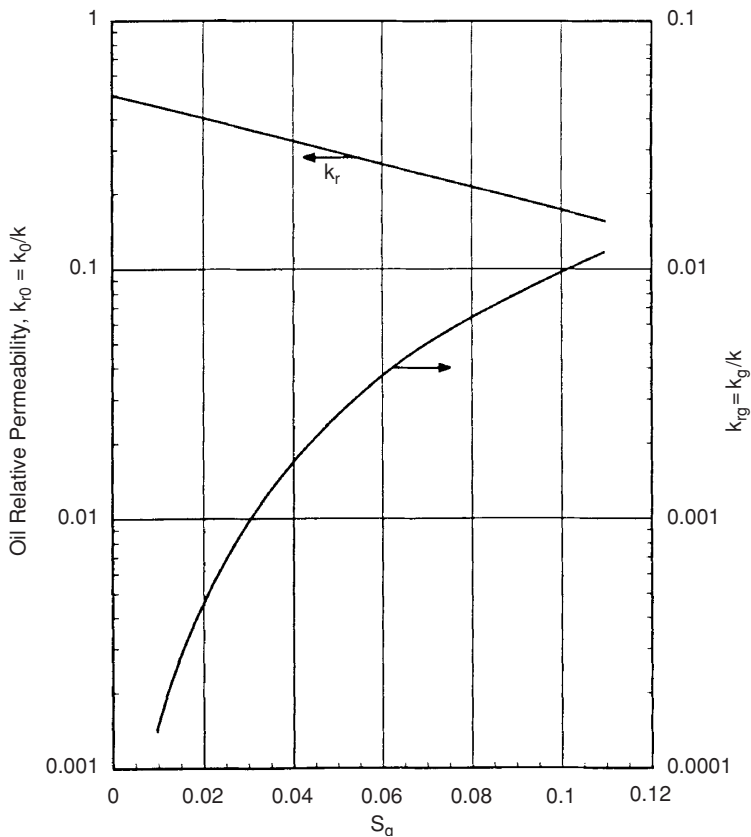


Figure 12-7. Relative permeability data for Example 12-3. (After Economides, M., et al., *Petroleum Production Systems*, Prentice Hall Petroleum Engineers Series, 1994.)

The following additional data are available:

$$p_i = p_b = 4350 \text{ psi} \quad S_{wi} = 30\% \quad N = 15 \text{ MMSTB}$$

Predict the cumulative oil and gas production to 3,350 psi.

Solution

A sample of the Tracy's calculation procedure is performed at 4,150 psi.

Step 1. Calculate Tracy's PVT functions at 4,150.

- Calculate the term Den from Equation 12-37

$$\text{Den} = (B_o - B_{oi}) + (R_{si} - R_s) B_g$$

$$\text{Den} = (1.42 - 1.43) + (840 - 820) (7.1 \times 10^{-4}) = 0.0042$$

- Calculate Φ_o and Φ_g

$$\Phi_o = (B_o - R_s B_g) / \text{Den}$$

$$\Phi_o = [1.42 - (820) (7.1 \times 10^{-4})] / 0.0042 = 199$$

$$\Phi_g = B_g / \text{Den}$$

$$= 7.1 \times 10^{-4} / 0.0042 = 0.17$$

Similarly, these PVT variables are calculated for all other pressures to give:

p	Φ_o	Φ_g
4350	—	—
4150	199	0.17
3950	49	0.044
3750	22.6	0.022
3550	13.6	0.014
3350	9.42	0.010

Step 2. Assume a value for the GOR at 4,150 psi as 850 scf/STB.

Step 3. Calculate the average GOR.

$$(\text{GOR})_{\text{avg}} = \frac{840 + 850}{2} = 845 \text{ scf/STB}$$

Step 4. Calculate the incremental cumulative oil production ΔN_p .

$$\Delta N_p = \frac{1 - 0}{199 + (845)(0.17)} = 0.00292 \text{ STB}$$

Step 5. Calculate the cumulative oil production N_p .

$$N_p = N_p^* + \Delta N_p$$

$$N_p = 0 + 0.00292 = 0.00292$$

Step 6. Calculate oil and gas saturations.

$$S_o = (1 - N_p) (B_o/B_{oi}) (1 - S_{wi})$$

$$S_o = (1 - 0.00292) (1.42/1.43) (1 - 0.3) = 0.693$$

$$S_g = 1 - S_{wi} - S_o$$

$$S_g = 1 - 0.3 - 0.693 = 0.007$$

Step 7. Determine the relative permeability ratio k_{rg}/k_{ro} from Figure 12-7, to give:

$$k_{rg}/k_{ro} = 8 \times 10^{-5}$$

Step 8. Using $\mu_o = 1.7$ cp and $\mu_g = 0.023$ cp, calculate the instantaneous GOR.

$$\text{GOR} = 820 + (1.7 \times 10^{-4}) \frac{(1.7)(1.42)}{(0.023)(7.1 \times 10^{-4})} = 845 \text{ scf/STB}$$

which agrees with the assumed value.

Step 9. Calculate cumulative gas production.

$$G_p = 0 + (0.00292) (850) = 2.48$$

Complete results of the method are shown below:

$\bar{p}$	ΔN_p	N_p	(GOR) _{avg}	ΔG_p	G_p scf/STB	$N_p = 15 \times 10^6$ N STB	$G_p = 15 \times 10^6$ N scf
4350	—	—	—	—	—	—	—
4150	0.00292	0.00292	845	2.48	2.48	0.0438×10^6	37.2×10^6
3950	0.00841	0.0110	880	7.23	9.71	0.165×10^6	145.65×10^6
3750	0.0120	0.0230	1000	12	21.71	0.180×10^6	325.65×10^6
3550	0.0126	0.0356	1280	16.1	37.81	0.534×10^6	567.15×10^6
3350	0.011	0.0460	1650	18.2	56.01	0.699×10^6	840×10^6

Muskat's Method

Muskat (1945) expressed the material balance equation for a depletion-drive reservoir in the following differential form:

$$\frac{dS_o}{dp} = \frac{\frac{S_o B_g}{B_o} \frac{dR_s}{dp} + \frac{S_o}{B_o} \frac{k_{rg}}{k_{ro}} \frac{\mu_o}{\mu_g} \frac{dB_o}{dp} + (1 - S_o - S_{wc}) B_g \frac{d(1/B_g)}{dp}}{1 + \frac{\mu_o}{\mu_g} \frac{k_{rg}}{k_{ro}}} \quad (12-45)$$

with

$$\Delta S_o = S_o^* - S_o$$

$$\Delta p = p^* - p$$

where S_o^* , p^* = oil saturation and average reservoir pressure at the beginning of the pressure step

S_o , p = oil saturation and average reservoir pressure at the end of the time step

R_s = gas solubility, scf/STB

B_g = gas formation volume factor, bbl/scf

Craft, Hawkins, and Terry (1991) suggested the calculations can be greatly facilitated by computing and preparing in advance in graphical form the following pressure dependent groups:

$$X(p) = \frac{B_g}{B_o} \frac{dR_s}{dp} \quad (12-46)$$

$$Y(p) = \frac{1}{B_o} \frac{\mu_o}{\mu_g} \frac{dB_o}{dp} \quad (12-47)$$

$$Z(p) = B_g \frac{d(1/B_g)}{dp} \quad (12-48)$$

Introducing the above pressure dependent terms into Equation 12-45, gives:

$$\left(\frac{\Delta S_o}{\Delta p} \right) = \frac{S_o X(p) + S_o \frac{k_{rg}}{k_{ro}} Y(p) + (1 - S_o - S_{wc}) Z(p)}{1 + \frac{\mu_o}{\mu_g} \frac{k_{rg}}{k_{ro}}} \quad (12-49)$$

Craft, Hawkins, and Terry (1991) proposed the following procedure for solving Muskat's equation for a given pressure drop Δp , i.e., $(p^* - p)$:

- Step 1.* Prepare a plot of k_{rg}/k_{ro} versus gas saturation.
- Step 2.* Plot R_s , B_o , and $(1/B_g)$ versus pressure and determine the slope of each plot at selected pressures, i.e., dB_o/dp , dR_s/dp , and $d(1/B_g)/dp$.
- Step 3.* Calculate the pressure dependent terms $X(p)$, $Y(p)$, and $Z(p)$ that correspond to the selected pressures in Step 2.
- Step 4.* Plot the pressure dependent terms as a function of pressure, as illustrated in Figure 12-8.
- Step 5.* Graphically determine the values of $X(p)$, $Y(p)$, and $Z(p)$ that correspond to the pressure p .
- Step 6.* Solve Equation 12-49 for $(\Delta S_o/\Delta p)$ by using the oil saturation S_o^* at the beginning of the pressure drop interval p^* .

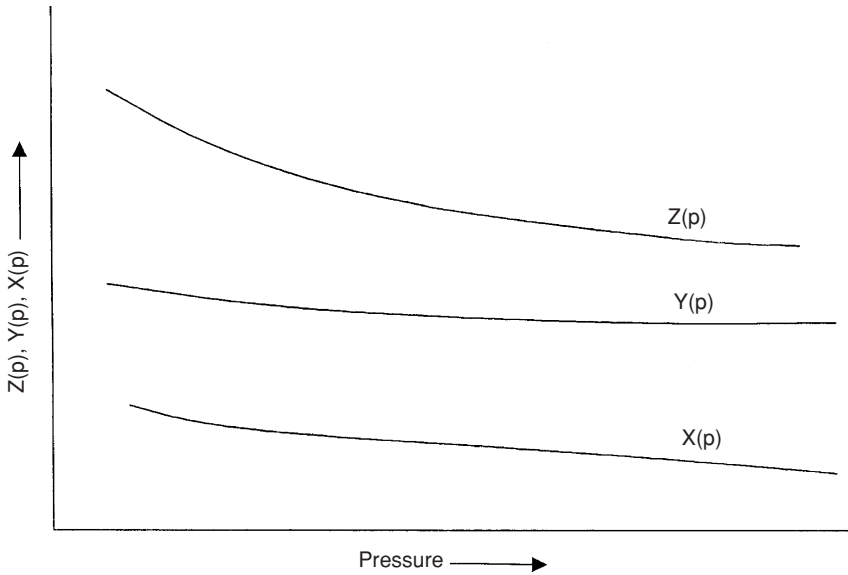


Figure 12-8. Pressure dependant terms vs. p .

Step 7. Determine the oil saturation S_o at the average reservoir pressure p , from:

$$S_o = S_o^* - (p^* - p) \left(\frac{\Delta S_o}{\Delta p} \right) \quad (12-50)$$

Step 8. Using the S_o from Step 7 and the pressure p , recalculate $(\Delta S_o / \Delta p)$ from Equation 12-49.

Step 9. Calculate the average value for $(\Delta S_o / \Delta p)$ from the two values obtained in Steps 6 and 8, or:

$$\left(\frac{\Delta S_o}{\Delta p} \right)_{\text{avg}} = \frac{1}{2} \left[\left(\frac{\Delta S_o}{\Delta p} \right)_{\text{step 6}} + \left(\frac{\Delta S_o}{\Delta p} \right)_{\text{step 8}} \right]$$

Step 10. Using $(\Delta S_o/\Delta p)_{\text{avg}}$, solve for the oil saturation S_o from:

$$S_o = S_o^* - (p^* - p) \left(\frac{\Delta S_o}{\Delta p} \right)_{\text{avg}} \quad (12-51)$$

This value of S_o becomes S_o^* for the next pressure drop interval.

Step 11. Calculate gas saturation S_g by:

$$S_g = 1 - S_{wi} - S_o$$

Step 12. Using the saturation equation, i.e., Equation 12-15, solve for the cumulative oil production.

$$N_P = N \left[1 - \left(\frac{B_{oi}}{B_o} \right) \left(\frac{S_o}{1 - S_{wi}} \right) \right] \quad (12-52)$$

Step 13. Calculate the cumulative gas production by using Equations 12-40 and 12-41.

Step 14. Repeat Steps 5 through 13 for all pressure drops of interest.

Example 12-5²

A volumetric depletion-drive reservoir exists at its bubble-point pressure of 2,500 psi. Detailed fluid property data are listed by Craft and his coauthors and given here at only two pressures.

Fluid Property	$p_b = 2500$ psi	$p = 2300$ psi
B_o , bbl/STB	1.498	1.463
R_s , scf/STB	721	669
B_g , bbl/scf	0.001048	0.001155
μ_o , cp	0.488	0.539
μ_g , cp	0.0170	0.0166
X (p)	0.00018	0.00021
Y (p)	0.00328	0.00380
Z (p)	0.00045	0.00050

²Craft, B. C., Hawkins, M., Terry, R., *Applied Petroleum Reservoir Engineering*, 2nd ed. Prentice Hall, 1991.

The following additional information is available:

$$N = 56 \text{ MMSTB} \quad S_{wi} = 20\% \quad S_{oi} = 80\%$$

S_g	k_{rg}/k_{ro}
0.10	0.010
0.20	0.065
0.30	0.200
0.50	2.000
0.55	3.000
0.57	5.000

Calculate the cumulative oil production for a pressure drop of 200 psi, i.e., at 2,300 psi.

Solution

Step 1. Using the oil saturation at the beginning of the pressure interval, i.e., $S_o^* = 0.8$, calculate (k_{rg}/k_{ro}) to give:

$$k_{rg}/k_{ro} = 0.0 \text{ (No free gas initially in place.)}$$

Step 2. Evaluate $(\Delta S_o/\Delta p)$ by applying Equation 12-49.

$$\left(\frac{\Delta S_o}{\Delta p} \right) = \frac{(0.8)(0.00018) + 0 + 0}{1 + 0} = 0.000146$$

Step 3. Estimate the oil saturation at $p = 2,300$ psi from Equation 12-51.

$$S_o = 0.8 - 200(0.000146) = 0.7709$$

Step 4. Recalculate $(\Delta S_o/\Delta p)$ by using $S_o = 0.7709$ and the pressure dependent terms at 2,300 psi.

$$\left(\frac{\Delta S_o}{\Delta p} \right) = \frac{0.7709(0.00021) + 0.7709(0.00001)0.0038 + (1 - 0.2 - 0.7709)0.0005}{1 + \left(\frac{0.539}{0.0166} \right)(0.00001)}$$

$$\left(\frac{\Delta S_o}{\Delta p} \right) = 0.000173$$

Step 5. Calculate the average $(\Delta S_o/\Delta p)$.

$$\left(\frac{\Delta S_o}{\Delta p} \right)_{\text{avg}} = \frac{0.000146 + 0.000173}{2} = 0.000159$$

Step 6. Calculate $S_o = 0.8 - (2500 - 2300)(0.000159) = 0.7682$.

Step 7. Calculate gas saturation.

$$S_g = 1 - 0.2 - 0.7682 + 0.0318$$

Step 8. Calculate cumulative oil production at 2,300 psi by using Equation 12-52.

$$N_p = 56 \times 10^6 \left[1 - \left(\frac{1.498}{1.463} \right) \left(\frac{0.7682}{1 - 0.2} \right) \right] = 939,500 \text{ STB}$$

Step 9. Calculate k_{rg}/k_{ro} at 2,300 psi, to give $k_{rg}/k_{ro} = 0.00001$.

Step 10. Calculate the instantaneous GOR at 2,300 psi.

$$\text{GOR} = 669 + 0.00001 \frac{(0.539)(1.463)}{(0.0166)(0.001155)} = 669 \text{ scf/STB}$$

Step 11. Calculate cumulative gas production.

$$G_p = \left(\frac{669 + 669}{2} \right) 939,500 = 629 \text{ MMscf}$$

It should be stressed that this method is based on the assumption of uniform oil saturation in the whole reservoir and that the solution will therefore break down when there is appreciable gas segregation in the formation. It is therefore applicable only when permeabilities are relatively low.

Tarner's Method

Tarner (1944) suggests an iterative technique for predicting cumulative oil production N_p and cumulative gas production G_p as a function of

reservoir pressure. The method is based on solving the material-balance equation and the instantaneous gas-oil ratio equation simultaneously for a given reservoir pressure drop from p_1 to p_2 . It is accordingly assumed that the cumulative oil and gas production has increased from N_{p1} and G_{p1} to N_{p2} and G_{p2} . To simplify the description of the proposed iterative procedure, the stepwise calculation is illustrated for a volumetric saturated-oil reservoir. It should be pointed out that Tarner's method could be extended to predict the volumetric behavior of reservoirs under different driving mechanisms.

Step 1. Select a future reservoir pressure p_2 below the initial (current) reservoir pressure p_1 and obtain the necessary PVT data. Assume that the cumulative oil production has increased from N_{p1} to N_{p2} . It should be pointed out that N_{p1} and G_{p1} are set equal to zero at the initial reservoir pressure, i.e., bubble-point pressure.

Step 2. Estimate or guess the cumulative oil production N_{p2} at p_2 .

Step 3. Calculate the cumulative gas production G_{p2} by rearranging the MBE, i.e., Equation 12-33, to give:

$$G_{p2} = N \left[(R_{si} - R_s) - \frac{B_{oi} - B_o}{B_g} \right] - N_{p2} \left[\frac{B_o}{B_g} - R_s \right] \quad (12-53)$$

Equivalently, the above relationship can be expressed in terms of the two-phase (total) formation volume factor B_t as:

$$G_{p2} = \frac{N(B_t - B_{ti}) - N_{p2}(B_t - R_{si}B_g)}{B_g} \quad (12-54)$$

where B_g = gas formation volume factor at p_2 , bbl/scf

B_o = oil formation volume factor at p_2 , bbl/STB

B_t = two-phase formation volume factor at p_2 , bbl/STB

N = initial oil-in-place, STB

Step 4. Calculate the oil and gas saturations at the assumed cumulative oil production N_{p2} and the selected reservoir pressure p_2 by applying Equations 12-15 and 12-16 respectively, or:

$$S_o = (1 - S_{wi}) \left[1 - \frac{N_{p2}}{N} \right] \left(\frac{B_o}{B_{oi}} \right)$$

$$S_g = 1 - S_o - S_{wi}$$

where B_o = initial oil formation volume factor at p_i , bbl/STB

B_o = oil formation volume factor at p_2 , bbl/STB

S_g = gas saturation at p_2

B_o = oil saturation at p_2

Step 5. Using the available relative permeability data, determine the relative permeability ratio k_{rg}/k_{ro} that corresponds to the gas saturation at p_2 and compute the instantaneous $(GOR)_2$ at p_2 from Equation 12-1, as:

$$(GOR)_2 = R_s + \left(\frac{k_{rg}}{k_{ro}} \right) \left(\frac{\mu_o B_o}{\mu_g B_g} \right) \quad (12-55)$$

It should be noted that all the PVT data in the expression must be evaluated at the assumed reservoir pressure p_2 .

Step 6. Calculate again the cumulative gas production G_{p2} at p_2 by applying Equation 12-7, or:

$$G_{p2} = (G_{p1}) + \left[\frac{(GOR)_1 + (GOR)_2}{2} \right] [N_{p2} - N_{p1}] \quad (12-56)$$

in which $(GOR)_1$ represents the instantaneous GOR at p_1 . If p_1 represents the initial reservoir pressure, then set $(GOR)_1 = R_{si}$.

Step 7. The total gas produced G_{p2} during the first prediction period as calculated by the material balance equation is compared to the total gas produced as calculated by the GOR equation. These two equations provide the two independent methods required for determining the total gas produced. Therefore, if the cumulative gas production G_{p2} as calculated from Step 3 agrees with the value of Step 6, the assumed value of N_{p2} is correct and a new

pressure may be selected and Steps 1 through 6 are repeated. Otherwise, assume another value of N_{p2} and repeat Steps 2 through 6.

Step 8. In order to simplify this iterative process, three values of N_p can be assumed, which yield three different solutions of cumulative gas production for each of the equations (i.e., MBE and GOR equation). When the computed values of G_{p2} are plotted versus the assumed values of N_{p2} , the resulting two curves (one representing results of Step 3 and the one representing Step 5) will intersect. This intersection indicates the cumulative oil and gas production that will satisfy both equations.

It should be pointed out that it may be more convenient to assume values of N_p as a fraction of the initial oil-in-place N . For instance, N_p could be assumed as 0.01 N , rather than as 10,000 STB. In this method, a true value of N is not required. Results of the calculations would be, therefore, in terms of STB of oil produced per STB of oil initially in place and scf of gas produced per STB of oil initially in place.

To illustrate the application of Turner's method, Cole (1969) presented the following example:

Example 12-6

A saturated-oil reservoir has a bubble-point pressure of 2,100 psi at 175°F. The initial reservoir pressure is 2,925 psi. The following data summarize the rock and fluid properties of the field:

Original oil-in-place = 10 MMSTB

Connate-water saturation = 15%

Porosity = 12%

$c_w = 3.6 \times 10^{-6} \text{ psi}^{-1}$

$c_f = 4.9 \times 10^{-6} \text{ psi}^{-1}$

Basic PVT Data

p, psi	B_o , bbl/STB	B_t , bbl/STB	R_s , scf/STB	B_g , bbl/scf	μ_o/μ_g
2925	1.429	1.429	1340	—	—
2100	1.480	1.480	1340	0.001283	34.1
1800	1.468	1.559	1280	0.001518	38.3
1500	1.440	1.792	1150	0.001853	42.4

Relative Permeability Ratio

S_o , %	k_{rg}/k_{ro}
81	0.018
76	0.063
60	0.85
50	3.35
40	10.2

Predict cumulative oil and gas production at 2,100, 1,800, and 1,500 psi.

Solution

The required calculations will be performed under the following two different driving mechanisms:

- During the reservoir pressure declines from the initial reservoir pressure of 2,925 to the bubble-point pressure of 2,100 psi, the reservoir is considered undersaturated and, therefore, the MBE can be used directly to calculate cumulative production without resorting to the iterative technique.
- For reservoir pressures below the bubble-point pressure, the reservoir is treated as a saturated-oil reservoir and Tarner's method may be applied.

Phase 1: Oil recovery prediction above the bubble-point pressure

Step 1. Arrange the MBE (Equation 11-32) and solve for the cumulative oil as:

$$N_p = \frac{N[E_o + E_{f,w}]}{B_o} \quad (12-57)$$

where

$$E_{f,w} = B_{oi} \left[\frac{c_w S_w + c_f}{1 - S_{wi}} \right] (p_i - p)$$

$$E_o = B_o - B_{oi}$$

Step 2. Calculate the two expansion factors E_o and $E_{f,w}$ for the pressure declines from 2,925 to 2,100 psi:

$$E_o = 1.480 - 1.429 = 0.051$$

$$E_{f,w} = 1.429 \left[\frac{(3.6 \times 10^{-6})(0.15) + (4.9 \times 10^{-6})}{1 - 0.15} \right] = 9.1456 \times 10^{-6}$$

Step 3. Calculate cumulative oil and gas production when the reservoir pressure declines from 2,925 to 2,100 psi by applying Equation 12-57, to give:

$$N_p = \frac{10 \times 10^6 [0.051 + 9.1456 \times 10^{-6}]}{1.48} = 344,656 \text{ STB}$$

At or above the bubble-point pressure, the producing gas-oil ratio is equal to the gas solubility at the bubble-point and, therefore, the cumulative gas production is given by:

$$G_p = N_p R_{si}$$

$$G_p = (344,656) (1340) = 462 \text{ MMscf}$$

Step 4. Determine remaining oil-in-place at 2,100 psi.

$$\text{Remaining oil-in-place} = 10,000,000 - 344,656 = 9,655,344 \text{ STB}$$

This remaining oil-in-place is considered as the initial oil-in-place during the reservoir performance below the saturation pressure, i.e.:

$$N = 9,655,344 \text{ STB}$$

$$N_p = 0.0 \text{ STB}$$

$$G_p = 0.0 \text{ scf}$$

$$R_{si} = 1340 \text{ scf/STB}$$

$$B_{oi} = 1.489 \text{ bbl/STB}$$

$$B_{ti} = 1.489 \text{ bbl/STB}$$

$$B_{gi} = 0.001283 \text{ bbl/scf}$$

Phase 2: Oil recovery prediction above the bubble-point pressure

First prediction period at 1,800 psi:

Step 1. Assume $N_p = 0.01$ N and apply Equation 12-54 to solve for G_p .

$$G_p = \frac{N(1.559 - 1.480) - (0.01N)(1.559 - 1340 \times 0.001518)}{0.001518} = 55.17 \text{ N}$$

Step 2. Calculate the oil saturation, to give:

$$S_o = (1 - S_{wi}) \left(1 - \frac{N_p}{N} \right) \frac{B_o}{B_{oi}} = (1 - 0.15) \left(1 - \frac{0.01N}{N} \right) \frac{1.468}{1.480} = 0.835$$

Step 3. Determine the relative permeability ratio k_{rg}/k_{ro} from the available data to give:

$$k_{rg}/k_{ro} = 0.0100$$

Step 4. Calculate the instantaneous GOR at 1,800 psi by applying Equation 12-55 to give:

$$GOR = 1280 + 0.0100(38.3) \left(\frac{1.468}{0.001518} \right) = 1650 \text{ scf/STB}$$

Step 5. Solve again for the cumulative gas production by using the average GOR and applying Equation 12-56 to yield:

$$G_p = 0 + \frac{1340 + 1650}{2} (0.01N - 0) = 14.95N$$

Step 6. Since the cumulative gas production as calculated by the two independent methods (Step 1 and Step 5) do not agree, the calculations must be repeated by assuming a different value for N_p and plotting results of the calculation. The final results as summarized below show the cumulative gas and oil production as the pressure declines from the bubble-point pressure. It should be pointed out that the cumulative production above the bubble-point pressure must be included when reporting the *total* cumulative oil and gas production.

Pressure	N_p	Actual N_p , STB	G_p	Actual G_p , MMscf
1800	0.0393 N	379,455	64.34 N	621.225
1500	0.0889 N	858,360	136.6 N	1318.92

PHASE 2. RELATING RESERVOIR PERFORMANCE TO TIME

All reservoir performance techniques show the relationship of cumulative oil production and the instantaneous GOR as a function of average reservoir pressure. These techniques, however, do not relate the cumulative oil production N_p and cumulative gas production G_p with time. Figure 12-9 shows a schematic illustration of the predicted cumulative oil production with reservoir pressure.

The time required for production can be calculated by applying the concept of the inflow performance relation (IPR) in conjunction with the

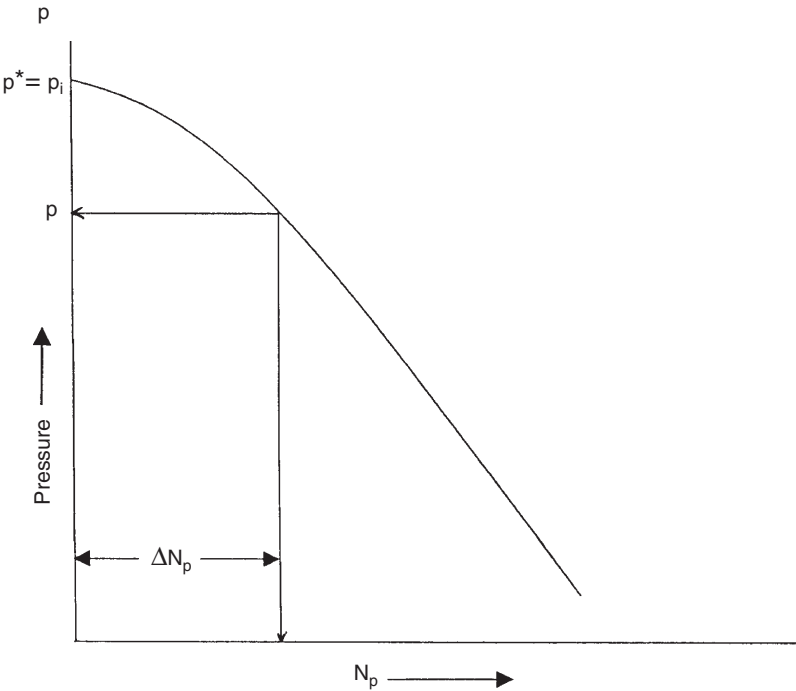
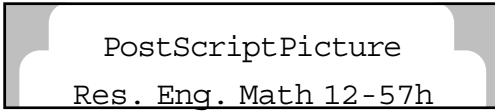


Figure 12-9. Cumulative production as a function of average reservoir pressure.

MBE predictions. Vogel (1969) expressed the well's inflow performance relationship by Equation 7-9, or:



The following methodology can be employed to correlate the predicted cumulative field production with time t .

Step 1. Plot the predicted cumulative oil production N_p as a function of average reservoir pressure p as shown in Figure 12-9.

Step 2. Construct the IPR curve for each well in field at the initial average reservoir pressure p^* . Calculate the oil flow rate for the entire field by taking the summation of the flow rates. Plot the flow rates as shown schematically in Figure 12-10 for two hypothetical wells and establish the IPR for the field.

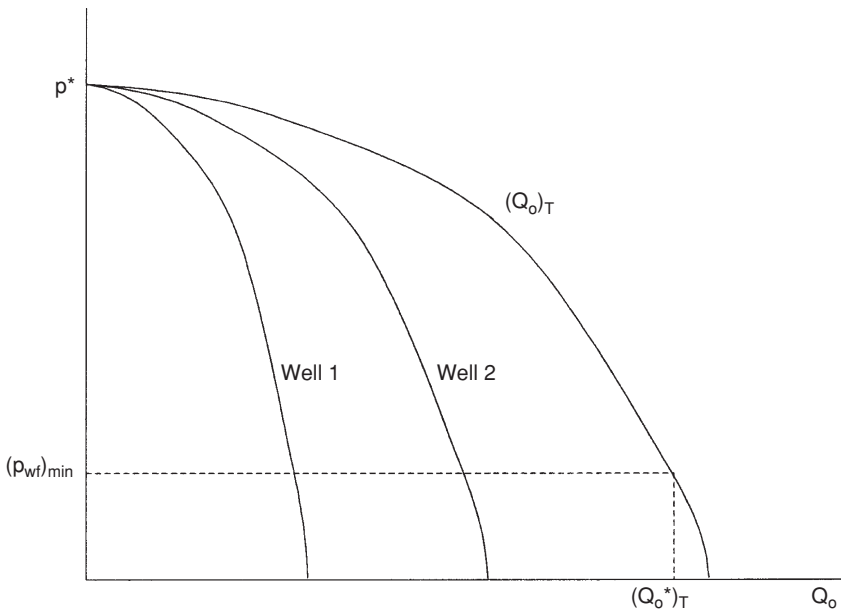


Figure 12-10. Overall field IPR at current pressure.

Step 3. Using the minimum bottom-hole flowing pressure $(p_{wf})_{min}$, determine the total field flow rate $(Q_o)_T^*$.

Step 4. Select a future average reservoir pressure p and determine the future IPR for each well in the field. Construct the field IPR curve as shown in Figure 12-11.

Step 5. Using the minimum p_{wf} , determine the field total oil flow rate $(Q_o)_T$.

Step 6. Calculate the average field production rate $(\bar{Q}_o)_T$.

$$(\bar{Q}_o)_T = \frac{(Q_o)_T + (Q_o)_T^*}{2}$$

Step 7. Calculate the time Δt required for the incremental oil production ΔN_p during the first pressure drop interval, i.e., from p^* to p , by:

$$\Delta t = \frac{\Delta N_p}{(\bar{Q}_o)_T}$$

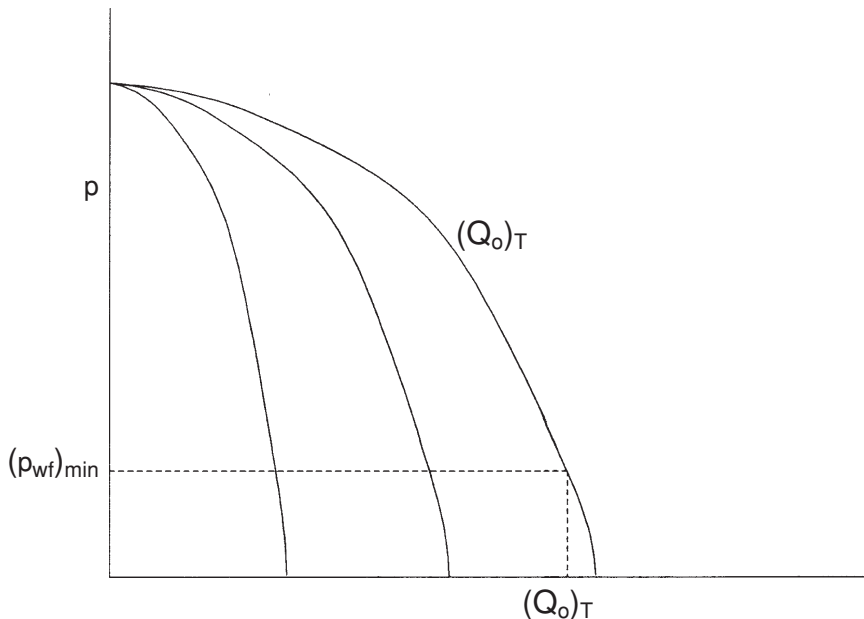


Figure 12-11. Future field IPR.

Step 8. Repeat the above steps and calculate the total time t to reach an average reservoir pressure p by:

$$t = \Sigma \Delta t$$

PROBLEMS

1. Determine the fractional oil recovery, during depletion down to bubble-point pressure, for the reservoir whose PVT parameters are listed in Table 3-7 and for which:

$$c_w = 3.5 \times 10^{-6} \text{ psi}^{-1} \quad c_f = 3.5 \times 10^{-6} \text{ psi}^{-1} \quad S_{wc} = 0.20$$

2. The Big Butte field is a depletion-drive reservoir that contains 25 MMSTB of oil initially in place. Tables 3-4 through 3-7 show the experimental PVT data of the crude oil system. The initial reservoir pressure is recorded as 1,936 psi at 247°F. The relative permeability ratio k_{rg}/k_{ro} is given by:

$$k_{rg}/k_{ro} = 0.007 e^{11.513 S_g}$$

Given

$$S_{or} = 35\% \quad S_{gc} = 3\% \quad S_{wi} = 25\%$$

and using a pressure drop increment of 200 psi, predict the reservoir future performance in terms of:

- Cumulative oil production N_p
- Cumulative gas production G_p
- Oil saturation S_o
- Gas saturation S_g
- Instantaneous GOR
- Cumulative producing gas-oil ratio R_p

Plot results of the calculations to an abandonment pressure of 500 psi. Use the following three methods:

1. Tracy's method
2. Muskat's method
3. Tarner's method

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